

Mid RC Part1

Comprehensive Notes on Set Theory, Functions, and Limits

Sincere gratitude to MATH1560J FA24 Teaching Team!

MATH1560J - Calculus

LiZi

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Exam Rules

What you should bring:

- 1 Student ID card
- 2 Stationery

What you shouldn't bring:

- 1 Textbook
- 2 Reference materials not provided officially
- 3 Electronic devices

Whether a calculator is permitted will be determined by the test content.

What we will provide:

- 1 Scratch paper
- 2 Reference including some derivative formulas
- 3 Clock
- 4 Pledge of Honor

If you have any questions:

- 1 Immediately ask TAs or the professor
- 2 No communication with anyone else

1 Set Theory Basics

1.1 Fundamental Concepts

Definition 1: Set

A **set** is an unordered collection of distinct objects. The objects in a set are called its **elements** or **members**.

- We write $a \in A$ if a is an element of the set A .
- We write $a \notin A$ if a is not an element of the set A .

Definition 2: Important Sets

- The **Empty Set**: The set containing no elements, denoted by $\emptyset$ or $\{\}$.
- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$: The set of Natural Numbers.
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$: The set of Integers.
- $\mathbb{Q}$: The set of Rational Numbers (ratios of integers).
- $\mathbb{R}$: The set of Real Numbers.
- $\mathbb{C}$: The set of Complex Numbers.

Definition 3: Cardinality

The **cardinality** of a set A , denoted $|A|$ or $\#A$, is its size (the number of elements it contains). If A is a finite set, $|A| = n$ for some $n \in \mathbb{N}$. Otherwise, the set is infinite and we write $|A| = \infty$.

1.2 Subsets and Intervals

Definition 4: Subset

A set A is a **subset** of a set B , written $A \subseteq B$, if every element of A is also an element of B .
A set A is a **proper subset** of B , written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Definition 5: Intervals

Given $a, b \in \mathbb{R}$:

- **Open Interval:** $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$
- **Closed Interval:** $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- **Half-Open Intervals:**
 - $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$
 - $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$

1.3 Set Operations

Definition 6: Cartesian Product

Given two sets A and B , the **Cartesian Product** $A \times B$ is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B := \{(a, b) \mid a \in A, b \in B\}$$

- For ordered pairs, order matters: $(a, b) = (x, y)$ if and only if $a = x$ and $b = y$.
- We can take the product of a set with itself: $A \times A = A^2$.
- This generalizes to n-tuples: $A^n = A \times \cdots \times A$ (n times).
- For example, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ represents the 2D Cartesian plane, and $\mathbb{R}^3$ represents 3D space.

Definition 7: Power Set

The **power set** of a set A , denoted $\mathcal{P}(A)$ or 2^A , is the set of all possible subsets of A .

Example 1: Power Set

If $A = \{1, 2\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

2 Foundations of Functions

2.1 Definition of a Function

A function (or map) is a fundamental concept that establishes a relationship between two sets.

Definition 8: Function

A function f consists of three essential parts:

1. **A domain (D):** The set of inputs for which the function is defined.
2. **A codomain (E):** The set of possible outputs.
3. **A rule of assignment:** A rule that assigns to *each* element x in the domain D *exactly one* element, denoted $f(x)$, in the codomain E .

We typically write this relationship as $f : D \rightarrow E$.

Definition 9: Image and Range

The **image** of a function $f : A \rightarrow B$ is the set of all actual output values:

$$\text{im}(f) = f(A) = \{f(x) \mid x \in A\}$$

The image $f(A)$ is a subset of the codomain B . The terms "image" and "range" are often used interchangeably.

Definition 10: Graph of a Function

For a function $f : D \rightarrow E$ where $D, E \subseteq \mathbb{R}$, the **graph** of f is the set of all ordered pairs $(x, f(x))$ in the Cartesian product $D \times E$.

$$\text{graph}(f) = \{(x, f(x)) \mid x \in D\} \subseteq \mathbb{R}^2$$

Property 1: The Vertical Line Test

A curve in the xy -plane represents the graph of a function of x if and only if no vertical line intersects the curve more than once. This test ensures that every input x has only one output $f(x)$.

2.2 Properties and Classifications of Functions

Definition 11: Parity

The parity of a function describes its symmetry.

- A function f is **even** if $f(-x) = f(x)$ for all x in its domain. The graph of an even function is symmetric about the y-axis.
- A function f is **odd** if $f(-x) = -f(x)$ for all x in its domain. The graph of an odd function has rotational symmetry about the origin.

Definition 12: Monotonicity

A function's monotonicity describes its directional behavior over an interval I .

- A function is **increasing** (or non-decreasing) on I if for any $x_1 < x_2$ in I , $f(x_1) \leq f(x_2)$. It is **strictly increasing** if $f(x_1) < f(x_2)$.
- A function is **decreasing** (or non-increasing) on I if for any $x_1 < x_2$ in I , $f(x_1) \geq f(x_2)$. It is **strictly decreasing** if $f(x_1) > f(x_2)$.

Definition 13: Mappings (Injective, Surjective, Bijective)

We can classify functions based on how their domain and codomain are mapped.

- **Injective (One-to-one):** A function $f : A \rightarrow B$ is injective if distinct elements in the domain map to distinct elements in the codomain.

$$\forall x, y \in A, x \neq y \implies f(x) \neq f(y)$$

(Equivalently: $f(x) = f(y) \implies x = y$)

- **Surjective (Onto):** A function $f : A \rightarrow B$ is surjective if its image equals its codomain (i.e., $f(A) = B$).

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

- **Bijective (One-to-one correspondence):** A function is bijective if it is both injective and surjective.

2.3 Common Types of Functions

A summary of elementary functions frequently encountered in calculus:

- **Polynomials:** Functions of the form $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$.
- **Power Functions:** Functions of the form $f(x) = x^a$, where a is a constant.

- **Rational Functions:** Ratios of two polynomials, $R(x) = P(x)/Q(x)$, defined where $Q(x) \neq 0$.
- **Trigonometric Functions:** $\sin$, $\cos$, $\tan$, and their reciprocals ($\csc$, $\sec$, $\cot$).
- **Exponential Functions:** Functions of the form $f(x) = a^x$, where the base a is a positive constant.
- **Logarithmic Functions:** The inverse of exponential functions, $f(x) = \log_a x$.
- **Hyperbolic Functions:** Combinations of exponential functions, such as $\sinh(x) = \frac{e^x - e^{-x}}{2}$ and $\cosh(x) = \frac{e^x + e^{-x}}{2}$.

2.4 Combining and Transforming Functions

Definition 14: Composition of Functions

Given two functions $f : A \rightarrow B$ and $g : B \rightarrow C$, the composite function $g \circ f$ (read "g of f") is defined as:

$$(g \circ f)(x) = g(f(x))$$

The domain of $g \circ f$ is A and its codomain is C ($g \circ f : A \rightarrow C$). Composition is associative, i.e., $h \circ (g \circ f) = (h \circ g) \circ f$, but not generally commutative ($f \circ g \neq g \circ f$).

Property 2: Function Transformations

Let $c > 0$ be a constant.

- **Vertical Shifts:** $y = f(x) + c$ (shift c units upward) and $y = f(x) - c$ (shift c units downward).
- **Horizontal Shifts:** $y = f(x - c)$ (shift c units to the right) and $y = f(x + c)$ (shift c units to the left).
- **Stretching/Shrinking:** $y = cf(x)$ (stretch vertically if $c > 1$, shrink if $0 < c < 1$) and $y = f(cx)$ (shrink horizontally if $c > 1$, stretch if $0 < c < 1$).
- **Reflections:** $y = -f(x)$ (reflect about the x-axis) and $y = f(-x)$ (reflect about the y-axis).



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$$x^2 = \underbrace{x + x + \cdots + x}_{(x \text{ times})}$$

$$\frac{d}{dx}x^2 = \frac{d}{dx} \left[\underbrace{x + x + \cdots + x}_{(x \text{ times})} \right]$$

$$2x = 1 + 1 + \cdots + 1 = x$$

$$2 = 1$$

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edited Apr 15, 2013 at 1:28

3 The Concept of a Limit

3.1 Intuitive Definition of a Limit

Definition 15: Limit of a Function

We write

$$\lim_{x \rightarrow a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a , equals L " if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a (on either side of a) but **not equal to a** .

Note 1: Important Note

The value of the limit at a point a does not depend on the value of the function *at* a . The function may even be undefined at a .

Definition 16: One-Sided Limits

- The **left-hand limit**, $\lim_{x \rightarrow a^-} f(x) = L$, means $f(x)$ approaches L as x approaches a from values less than a .
- The **right-hand limit**, $\lim_{x \rightarrow a^+} f(x) = L$, means $f(x)$ approaches L as x approaches a from values greater than a .

The two-sided limit $\lim_{x \rightarrow a} f(x) = L$ exists if and only if both one-sided limits exist and are equal: $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$.

3.2 Infinite Limits and Asymptotes

Definition 17: Infinite Limits

The expression $\lim_{x \rightarrow a} f(x) = \infty$ means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a (but not equal to a). Similarly, $\lim_{x \rightarrow a} f(x) = -\infty$ means the values can be made arbitrarily large and negative.

Note 2: Warning

Saying a limit is ∞ or $-\infty$ is a specific way of saying the limit **does not exist** as a finite number. **In the mid term exam, if you get the limit of a function is ∞ or $-\infty$, you should clarify that the limit **does not exist (DNE)**, otherwise we may give you some deductions!**

Definition 18: Vertical Asymptote

The line $x = a$ is a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following one-sided limits is infinite:

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty$$

This typically occurs for rational functions at the zeros of the denominator.

Definition 19: Limits at Infinity

The expression $\lim_{x \rightarrow \infty} f(x) = L$ means that the values of $f(x)$ can be made arbitrarily close to L by taking x to be sufficiently large. A similar definition holds for $x \rightarrow -\infty$.

Definition 20: Horizontal Asymptote

The line $y = L$ is a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Example 2: Asymptotes of $\arctan(x)$

The function $f(x) = \arctan(x)$ has two horizontal asymptotes:

- $y = \frac{\pi}{2}$, because $\lim_{x \rightarrow \infty} \arctan(x) = \frac{\pi}{2}$.
- $y = -\frac{\pi}{2}$, because $\lim_{x \rightarrow -\infty} \arctan(x) = -\frac{\pi}{2}$.

4 Limit Laws and Calculation Techniques

Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist as finite numbers.

4.1 The Limit Laws

Law 1: Basic Limit Laws

1. **Sum/Difference Rule:** $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$.
2. **Constant Multiple Rule:** $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$.
3. **Product Rule:** $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = (\lim_{x \rightarrow a} f(x)) \cdot (\lim_{x \rightarrow a} g(x))$.
4. **Quotient Rule:** $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$, provided $\lim_{x \rightarrow a} g(x) \neq 0$.
5. **Power Rule:** $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$, where n is a positive integer.
6. **Root Rule:** $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$, where n is a positive integer. If n is even, we assume $\lim_{x \rightarrow a} f(x) > 0$.

Example 3: Example (Limit Laws)

Evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) \left(2 - \frac{1}{x^2}\right)$.

Solution: Using the product rule for limits, and knowing $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ and $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) \cdot \lim_{x \rightarrow \infty} \left(2 - \frac{1}{x^2}\right) = (1 + 0)(2 - 0) = 2$$

Property 3: Direct Substitution

If f is a polynomial or a rational function and a is in the domain of f , then $\lim_{x \rightarrow a} f(x) = f(a)$.

4.2 Techniques for Indeterminate Forms

Often, direct substitution leads to indeterminate forms like $\frac{0}{0}$ or $\frac{\infty}{\infty}$. We use various techniques to resolve these.

Rule 1: Algebraic Manipulation

For indeterminate forms, try algebraic simplification such as factoring and canceling common terms, or rationalizing the numerator or denominator.

Example 4: Example (Factorization)

Evaluate $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.

Solution: This is of the form $\frac{0}{0}$. We factor the denominator:

$$\lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{1+1} = \frac{1}{2}$$

Example 5: Example (Rationalization)

Evaluate $\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9}-3}{t^2}$.

Solution: This is of type $\frac{0}{0}$. We rationalize the numerator:

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\sqrt{t^2+9}-3}{t^2} &= \lim_{t \rightarrow 0} \left[\frac{\sqrt{t^2+9}-3}{t^2} \cdot \frac{\sqrt{t^2+9}+3}{\sqrt{t^2+9}+3} \right] \\ &= \lim_{t \rightarrow 0} \frac{(t^2+9)-9}{t^2(\sqrt{t^2+9}+3)} \\ &= \lim_{t \rightarrow 0} \frac{t^2}{t^2(\sqrt{t^2+9}+3)} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2+9}+3} \\ &= \frac{1}{\sqrt{0+9}+3} = \frac{1}{3+3} = \frac{1}{6} \end{aligned}$$

Theorem 1: The Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ for all x near a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then $\lim_{x \rightarrow a} g(x) = L$.

Example 6: Example (Squeeze Theorem)

Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$.

Solution: We know that $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$ for all $x \neq 0$. Multiplying by x^2 (which is non-negative) gives:

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

Since $\lim_{x \rightarrow 0}(-x^2) = 0$ and $\lim_{x \rightarrow 0} x^2 = 0$, by the Squeeze Theorem, we conclude:

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

Rule 2: L'Hôpital's Rule

Suppose f and g are differentiable and we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ as $x \rightarrow a$. Then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right side exists (or is $\pm\infty$). This rule is also valid for one-sided limits and limits at infinity.

Example 7: Example (L'Hôpital's $\frac{0}{0}$)

Evaluate $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$.

Solution: This is of type $\frac{0}{0}$.

$$\lim_{x \rightarrow 1} \frac{\ln x}{x-1} \stackrel{L'H}{=} \lim_{x \rightarrow 1} \frac{1/x}{1} = \frac{1/1}{1} = 1$$

Example 8: Example (L'Hôpital's $\frac{\infty}{\infty}$)

Evaluate $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

Solution: This is of type $\frac{\infty}{\infty}$. We apply the rule twice.

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$$

(The limit does not exist as a finite number).

Example 9: Example (Indeterminate Product $0 \cdot \infty$)

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Solution: This is of type $0 \cdot (-\infty)$. We rewrite it as a quotient (type $\frac{\infty}{\infty}$):

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

Example 10: Example (Indeterminate Form $\infty - \infty$)

Evaluate $\lim_{x \rightarrow (\pi/2)^-} (\sec(x) - \tan(x))$.

Solution: This is of type $\infty - \infty$. We rewrite it as a single quotient:

$$\lim_{x \rightarrow (\pi/2)^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) = \lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x}$$

This is now of the form $\frac{0}{0}$. Applying L'Hôpital's Rule:

$$\stackrel{L'H}{=} \lim_{x \rightarrow (\pi/2)^-} \frac{-\cos x}{-\sin x} = \frac{-\cos(\pi/2)}{-\sin(\pi/2)} = \frac{0}{-1} = 0$$

Example 11: Example (Indeterminate Power 0^0)

To evaluate $\lim_{x \rightarrow 0^+} x^x$.

Solution: This is of type 0^0 . Let $y = x^x$. Then $\ln y = x \ln x$. From the previous example, we know $\lim_{x \rightarrow 0^+} x \ln x = 0$. Since the exponential function is continuous, we can write:

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{\ln y} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = 1$$

Example 12: Example (Indeterminate Power 1^∞)

Evaluate $\lim_{x \rightarrow \infty} (1 - \frac{1}{x})^{kx}$ where k is a positive integer.

Solution: This is of type 1^∞ . Let $y = (1 - \frac{1}{x})^{kx}$.

$$\ln y = kx \ln \left(1 - \frac{1}{x} \right) = k \frac{\ln(1 - 1/x)}{1/x}$$

Let $u = 1/x$. As $x \rightarrow \infty$, $u \rightarrow 0^+$.

$$\lim_{x \rightarrow \infty} \ln y = k \lim_{u \rightarrow 0^+} \frac{\ln(1 - u)}{u}$$

This is of type $\frac{0}{0}$. Using L'Hôpital's Rule:

$$\stackrel{L'H}{=} k \lim_{u \rightarrow 0^+} \frac{\frac{-1}{1-u}}{1} = k \left(\frac{-1}{1-0} \right) = -k$$

Since $\lim \ln y = -k$, the original limit is e^{-k} .

4.3 Important Limits and Equivalent Infinitesimals

These are useful shortcuts for limit calculations.

Property 4: List of Important Limits

- (a) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- (b) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$
- (c) $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$
- (d) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- (e) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- (f) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- (g) $\lim_{x \rightarrow 0^+} x^x = 1$

Definition 21: Equivalent Infinitesimals

If $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 1$, we say $f(x)$ is equivalent to $g(x)$ as $x \rightarrow 0$, written $f(x) \sim g(x)$. We can substitute equivalent infinitesimals in products and quotients to simplify limits.

Common Equivalences as $x \rightarrow 0$:

- $\sin x \sim x$
- $\tan x \sim x$
- $1 - \cos x \sim \frac{x^2}{2}$
- $\ln(1 + x) \sim x$
- $e^x - 1 \sim x$
- $\arcsin x \sim x$
- $\arctan x \sim x$
- $(1 + x)^k - 1 \sim kx$

Example 13: Example (Equivalent Infinitesimals)

Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+4x)}{\sin(3x)}$.

Solution: Using infinitesimals, as $x \rightarrow 0$, we have $\ln(1 + 4x) \sim 4x$ and $\sin(3x) \sim 3x$.

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 4x)}{\sin(3x)} = \lim_{x \rightarrow 0} \frac{4x}{3x} = \frac{4}{3}$$

Example 14: Example (L'Hôpital's or Definition of Derivative)

Evaluate $\lim_{x \rightarrow 0} \frac{-1 + \sqrt[n]{x+1}}{x}$.

Solution: This is of type $\frac{0}{0}$. We can use L'Hôpital's Rule. Let $f(x) = (x + 1)^{1/n}$. Then $f'(x) = \frac{1}{n}(x + 1)^{\frac{1}{n}-1}$.

$$\lim_{x \rightarrow 0} \frac{(x + 1)^{1/n} - 1}{x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{n}(x + 1)^{\frac{1}{n}-1}}{1} = \frac{1}{n}(0 + 1)^{\frac{1}{n}-1} = \frac{1}{n}$$

5 Continuity

5.1 Definition of Continuity

Definition 22: Continuity at a Point

A function f is **continuous** at a number a if:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

This definition implicitly requires three conditions:

1. $f(a)$ is defined (i.e., a is in the domain of f).
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. The limit equals the function value: $\lim_{x \rightarrow a} f(x) = f(a)$.

If a function is not continuous at a , it has a **discontinuity** at a .

Definition 23: Continuity on an Interval

A function is continuous on an interval if it is continuous at every number in the interval. For endpoints, we check for one-sided continuity (continuity from the right or left).

5.2 Properties of Continuous Functions

Theorem 2: Continuity Laws

If f and g are continuous at a , then the following are also continuous at a :

1. $f + g$ and $f - g$
2. $c \cdot f$ for any constant c
3. $f \cdot g$
4. f/g , provided $g(a) \neq 0$

Theorem 3: Continuity of Composite Functions

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ is continuous at a . This allows us to "pass the limit inside" a continuous function:

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

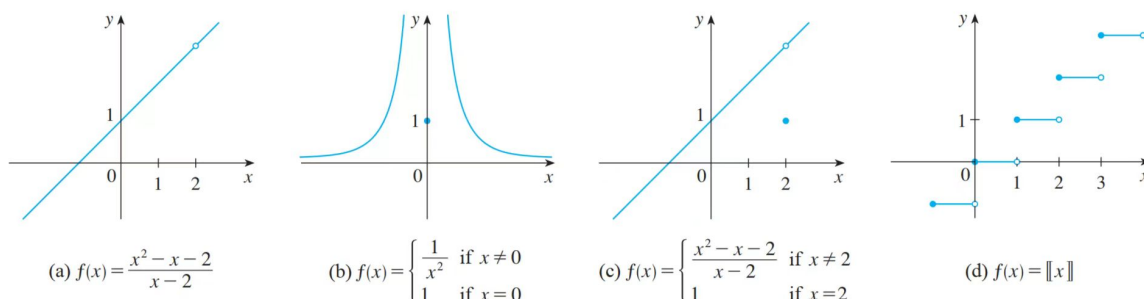
Property 5: Continuity of Common Functions

Polynomials, rational functions, root functions, trigonometric and inverse trigonometric functions, exponential functions, and logarithmic functions are all continuous at every number in their domains.

5.3 Types of Discontinuities

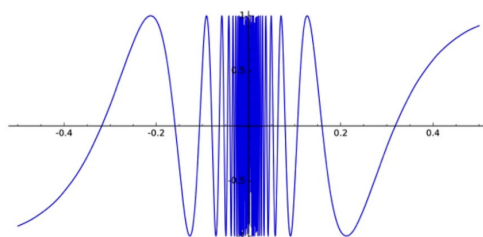
- **Removable Discontinuity:** Occurs when $\lim_{x \rightarrow a} f(x)$ exists but is not equal to $f(a)$ (either because $f(a)$ is different or is undefined).
- **Jump Discontinuity:** Occurs when the left-hand and right-hand limits exist but are not equal ($\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$).
- **Infinite Discontinuity:** Occurs when one or both of the one-sided limits are infinite. This is associated with a vertical asymptote.
- **Oscillating Discontinuity:** Occurs when the function oscillates too much to approach a single value, like $f(x) = \sin(1/x)$ at $x = 0$.

Figure 3 shows the graphs of the functions in Example 2. In each case the graph can't be drawn without lifting the pen from the paper because a hole or break or jump occurs in the graph. The kind of discontinuity illustrated in parts (a) and (c) is called **removable** because we could remove the discontinuity by redefining f at just the single number 2. [The function $g(x) = x + 1$ is continuous.] The discontinuity in part (b) is called an **infinite discontinuity**. The discontinuities in part (d) are called **jump discontinuities** because the function “jumps” from one value to another.



Continuity at a Point

Types of Discontinuity: Oscillating Discontinuity



6 Relationship: Limits, Continuity, Differentiability

The concepts of limits, continuity, and differentiability are hierarchically related.

Theorem 4: Differentiability Implies Continuity

If a function f is differentiable at a point a , then it must also be continuous at a .

Proof. We want to show that if f is differentiable at a , then f is continuous at a .

1. Definitions:

- **Differentiable at a (Given):** The limit $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists and equals a finite number $f'(a)$.
- **Continuous at a (To Prove):** We must show that $\lim_{x \rightarrow a} f(x) = f(a)$. This is equivalent to showing that $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$.

2. Proof: We start with the expression $f(x) - f(a)$ and use an algebraic manipulation. For $x \neq a$:

$$f(x) - f(a) = \left(\frac{f(x) - f(a)}{x - a} \right) \cdot (x - a)$$

Now, we take the limit of both sides as x approaches a :

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \lim_{x \rightarrow a} \left[\left(\frac{f(x) - f(a)}{x - a} \right) \cdot (x - a) \right]$$

By the product rule for limits, since both limits on the right exist:

$$= \left[\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right] \cdot \left[\lim_{x \rightarrow a} (x - a) \right]$$

We substitute the known values of these limits:

$$\begin{aligned} &= (f'(a)) \cdot (0) \\ &= 0 \end{aligned}$$

Since $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$, it follows that $\lim_{x \rightarrow a} f(x) = f(a)$. This is the definition of continuity at a . $\square$

Note 3: Converse is Not True

A function can be continuous at a point without being differentiable there. Common examples include:

- Functions with "corners" or "cusps", like $f(x) = |x|$ at $x = 0$.
- Functions with a vertical tangent line, like $f(x) = \sqrt[3]{x}$ at $x = 0$.

Remark 1: Summary of Relationships

- The existence of a **limit** at a point a is the most fundamental requirement. It describes the behavior of the function *near* a .
- **Continuity** is a stronger condition. It requires not only that the limit exists, but also that the function is defined at a and that the limit equals the function's value at a .
- **Differentiability** is the strongest condition. It requires the function to be "smooth" at a point, without any sharp corners, cusps, or vertical tangents. This requires the existence of a specific limit, the derivative, which in turn implies continuity.

In short: **Differentiability** $\implies$ **Continuity** $\implies$ **Limit Exists**.

7 Key Theorems on Continuous and Differentiable Functions

7.1 The Intermediate Value Theorem (IVT)

Theorem 5: Intermediate Value Theorem (IVT)

Suppose that f is continuous on a closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists at least one number c in the open interval (a, b) such that $f(c) = N$.

Remark 2: Significance of IVT

The IVT guarantees that a continuous function takes on every value between its values at the endpoints. It is a powerful tool for proving the existence of roots of equations.

Example 15: Example (Application of IVT - Fixed Point)

Prove that if $f : [0, 1] \rightarrow [0, 1]$ is continuous, there exists a point $c \in [0, 1]$ such that $f(c) = c$.

Proof. 1. Define a new function $g(x) = f(x) - x$.

2. Since $f(x)$ and x are continuous functions, $g(x)$ is also continuous on $[0, 1]$.

3. We evaluate $g(x)$ at the endpoints:

- $g(0) = f(0) - 0 = f(0)$. Since the range of f is $[0, 1]$, we know $f(0) \geq 0$.
- $g(1) = f(1) - 1$. Since the range of f is $[0, 1]$, we know $f(1) \leq 1$, so $f(1) - 1 \leq 0$.

4. We have $g(0) \geq 0$ and $g(1) \leq 0$. We are looking for a c such that $g(c) = 0$.

5. **Case 1:** If $g(0) = 0$, then $c = 0$ is our point ($f(0) = 0$).

6. **Case 2:** If $g(1) = 0$, then $c = 1$ is our point ($f(1) = 1$).

7. **Case 3:** If $g(0) > 0$ and $g(1) < 0$, then $N = 0$ is a number between $g(0)$ and $g(1)$. By the Intermediate Value Theorem, there must be some $c \in (0, 1)$ such that $g(c) = 0$.

8. In all cases, there exists a $c \in [0, 1]$ where $g(c) = 0$, which means $f(c) - c = 0$, or $f(c) = c$. □

7.2 Theorems on Differentiable Functions

These theorems are deeply connected to the properties of continuous functions over intervals.

Theorem 6: Darboux's Theorem (Intermediate Value Property of Derivatives)

If a function f is differentiable on an interval $[a, b]$, then its derivative f' has the intermediate value property. That is, for any value k between $f'(a)$ and $f'(b)$, there exists a $c \in (a, b)$ such that $f'(c) = k$.

Proof. (Based on lecture notes) Without loss of generality, assume $f'(a) < k < f'(b)$.

1. Define a new function $g(x) = f(x) - kx$ on $[a, b]$. Since f and kx are continuous and differentiable, so is g .

2. The derivative of g is $g'(x) = f'(x) - k$.

3. At the endpoints, we have:

- $g'(a) = f'(a) - k < 0$
- $g'(b) = f'(b) - k > 0$

4. Since g is continuous on the closed interval $[a, b]$, by the Extreme Value Theorem (which applies to continuous functions), g must attain a global minimum on this interval.

5. Since $g'(a) < 0$, for some values of x just to the right of a , $g(x) < g(a)$. Thus, the minimum cannot occur at $x = a$.

6. Since $g'(b) > 0$, for some values of x just to the left of b , $g(x) < g(b)$. Thus, the minimum cannot occur at $x = b$.

7. Therefore, the global minimum must occur at some point c in the open interval (a, b) .

8. By Fermat's Theorem, since g has a local (and global) minimum at c and is differentiable there, its derivative must be zero: $g'(c) = 0$.

9. This implies $f'(c) - k = 0$, or $f'(c) = k$. □

Note 4: Important Note

This is remarkable because f' itself is not required to be continuous for it to have the intermediate value property.

Theorem 7: Rolle's Theorem

Let f be a function that satisfies:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$.

Then there is a number c in (a, b) such that $f'(c) = 0$.

Remark 3: Interpretation of Rolle's Theorem

Rolle's Theorem states that if a differentiable function starts and ends at the same height, its derivative must be zero somewhere in between (i.e., there is a horizontal tangent).

Theorem 8: The Mean Value Theorem (MVT)

Let f be a function that satisfies:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. (Based on lecture notes) 1. We can prove this using Rolle's Theorem. Define an auxiliary function $h(x)$ that represents the vertical distance between the graph of $f(x)$ and the secant line connecting $(a, f(a))$ and $(b, f(b))$.

2. The equation of the secant line is $y = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$.

3. Let $h(x) = f(x) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(x-a) \right]$.

4. We check the conditions for Rolle's Theorem on $h(x)$:

- $h(x)$ is continuous on $[a, b]$ because it is the difference of continuous functions ($f(x)$ and a line).
- $h(x)$ is differentiable on (a, b) for the same reason.
- $h(a) = f(a) - [f(a) + 0] = 0$.
- $h(b) = f(b) - \left[f(a) + \frac{f(b)-f(a)}{b-a}(b-a) \right] = f(b) - [f(a) + f(b) - f(a)] = 0$.

5. Since $h(a) = h(b) = 0$, $h(x)$ satisfies all conditions for Rolle's Theorem. Therefore, there exists a $c \in (a, b)$ such that $h'(c) = 0$.

6. The derivative of $h(x)$ is $h'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$.

7. Setting $h'(c) = 0$ gives $f'(c) - \frac{f(b)-f(a)}{b-a} = 0$, which rearranges to the desired result: $f'(c) = \frac{f(b)-f(a)}{b-a}$. □

Remark 4: Interpretation of MVT

The MVT is a generalization of Rolle's Theorem. It states that for any differentiable function, there is a point c where the instantaneous rate of change ($f'(c)$) is equal to the average rate of change over the interval $[a, b]$. Geometrically, this means there is a tangent line parallel to the secant line connecting the endpoints.

8 More Exercises

Example 16

Suppose f is continuous on $[1, 5]$ and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, explain why $f(3) > 6$.

Example 17

Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)} = 0$.

Example 18

Suppose we know that $f(x)$ is continuous and differentiable on the interval $[-6, -1]$, that $f(-6) = -23$ and that $f'(x) \geq -4$. What is the smallest possible value for $f(-1)$?

Example 19

Evaluate the following limit:

$$\lim_{x \rightarrow 0^+} [\cos(2x)]^{1/x^2}$$

Example 20

Evaluate the following limit:

$$\lim_{t \rightarrow \infty} \left[t \ln \left(1 + \frac{3}{t} \right) \right]$$

Example 21

Evaluate $\lim_{x \rightarrow 0} \frac{|3x+4|-|x|-4}{x}$.

8.1 Solutions to Exercises

Solution to Example 16

This is a direct application of the Intermediate Value Theorem (IVT).

1. **Analyze the Interval:** We are interested in the interval $(1, 4)$. Since f is continuous on $[1, 5]$, it is also continuous on the open interval $(1, 4)$.
2. **Analyze the Function's Behavior:** We are told the *only* solutions to $f(x) = 6$ are at the endpoints $x = 1$ and $x = 4$. This means for any x in the open interval $(1, 4)$, we must have $f(x) \neq 6$.
3. **Apply the IVT Corollary:** A corollary of the IVT states that if a function is continuous on an interval and never equals a certain value k , then the function must either be *always greater than k* or *always less than k* on that interval.
4. **Use the Given Point:** We are given that $f(2) = 8$. Since $x = 2$ is in the interval $(1, 4)$ and $f(2) = 8 > 6$, this rules out the possibility that $f(x)$ is always less than 6.
5. **Conclude:** Therefore, the only possibility is that $f(x) > 6$ for all x in the interval $(1, 4)$. Since $3 \in (1, 4)$, it must be that $f(3) > 6$.

Solution to Example 17

We will prove this limit using the Squeeze Theorem.

1. **Establish a Bound:** The sine function is always bounded between -1 and 1, regardless of its input. For all $x \neq 0$:

$$-1 \leq \sin(\pi/x) \leq 1$$

2. **Bound the Exponential Part:** Since the exponential function $f(u) = e^u$ is an increasing function, we can apply it to all parts of the inequality:

$$e^{-1} \leq e^{\sin(\pi/x)} \leq e^1$$

This shows that the $e^{\sin(\pi/x)}$ part is bounded between $1/e$ and e .

3. **Apply the Squeezing Factor:** We are interested in the limit as $x \rightarrow 0^+$. For $x > 0$, $\sqrt{x}$ is positive, so we can multiply the inequality by $\sqrt{x}$ without changing the direction of the symbols:

$$\begin{aligned}\sqrt{x} \cdot e^{-1} &\leq \sqrt{x} e^{\sin(\pi/x)} \leq \sqrt{x} \cdot e^1 \\ \frac{\sqrt{x}}{e} &\leq \sqrt{x} e^{\sin(\pi/x)} \leq e\sqrt{x}\end{aligned}$$

4. **Apply the Squeeze Theorem:** Now we take the limit of the "lower bound" and "upper bound" functions as $x \rightarrow 0^+$.

- **Lower Bound Limit:** $\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e} = \frac{0}{e} = 0.$
- **Upper Bound Limit:** $\lim_{x \rightarrow 0^+} e\sqrt{x} = e \cdot 0 = 0.$

5. **Conclusion:** Since our function is squeezed between two functions that both approach 0, by the Squeeze Theorem, our function's limit must also be 0.

Solution to Example 18

We can solve this using the Mean Value Theorem (MVT).

1. **Check Conditions:** The problem states that $f(x)$ is continuous on $[-6, -1]$ and differentiable on the interval, so the conditions for the MVT are met.
2. **Apply the MVT:** The MVT guarantees that there exists at least one number c in the open interval $(-6, -1)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

3. **Substitute Values:** Let $a = -6$ and $b = -1$.

$$f'(c) = \frac{f(-1) - f(-6)}{(-1) - (-6)} = \frac{f(-1) - (-23)}{-1 + 6} = \frac{f(-1) + 23}{5}$$

4. **Use the Derivative Constraint:** We are given that $f'(x) \geq -4$ for all x in the interval. This must also be true for our specific c , so $f'(c) \geq -4$.
5. **Form an Inequality:** We substitute our expression for $f'(c)$ into the inequality:

$$\frac{f(-1) + 23}{5} \geq -4$$

6. **Solve for $f(-1)$:**

$$f(-1) + 23 \geq -20$$

$$f(-1) \geq -43$$

7. **Conclusion:** The inequality shows that the value of $f(-1)$ must be greater than or equal to -43. Therefore, the smallest possible value for $f(-1)$ is -43.

Solution to Example 19

This limit is of the indeterminate form 1^∞ . We can solve this using logarithms and L'Hôpital's Rule.

1. **Set up with Logarithms:** Let $L = \lim_{x \rightarrow 0^+} [\cos(2x)]^{1/x^2}$. Then, $\ln(L) = \lim_{x \rightarrow 0^+} \ln([\cos(2x)]^{1/x^2}) = \lim_{x \rightarrow 0^+} \frac{\ln(\cos(2x))}{x^2}$.
2. **Check Form:** As $x \rightarrow 0^+$, the numerator $\ln(\cos(0)) \rightarrow \ln(1) = 0$, and the denominator $x^2 \rightarrow 0$. This is an indeterminate form of type $\frac{0}{0}$.
3. **Apply L'Hôpital's Rule:** We differentiate the numerator and the denominator:

$$\begin{aligned} \ln(L) &= \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln(\cos(2x)))}{\frac{d}{dx}(x^2)} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{\cos(2x)} \cdot (-\sin(2x) \cdot 2)}{2x} \\ &= \lim_{x \rightarrow 0^+} \frac{-2 \sin(2x)}{2x \cos(2x)} = \lim_{x \rightarrow 0^+} \frac{-\tan(2x)}{x} \end{aligned}$$

4. **Apply L'Hôpital's Rule Again:** This is still a $\frac{0}{0}$ form.

$$\ln(L) = \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(-\tan(2x))}{\frac{d}{dx}(x)} = \lim_{x \rightarrow 0^+} \frac{-\sec^2(2x) \cdot 2}{1}$$

5. **Evaluate the Limit of $\ln(L)$:**

$$\ln(L) = \lim_{x \rightarrow 0^+} -2 \sec^2(2x) = -2 \cdot \sec^2(0) = -2 \cdot (1)^2 = -2$$

6. **Solve for the Original Limit L :** We found that $\lim_{x \rightarrow 0^+} \ln(L) = -2$.

$$L = e^{\lim_{x \rightarrow 0^+} \ln(L)} = e^{-2}$$

7. **Conclusion:** The limit is e^{-2} or $\frac{1}{e^2}$.

Solution to Example 20

This limit is of the indeterminate form $\infty \cdot 0$. We can solve this by rewriting it as a quotient and using L'Hôpital's Rule.

1. **Rewrite as a Quotient:** We rewrite the expression as a fraction to get a $\frac{0}{0}$ form:

$$\lim_{t \rightarrow \infty} \left[t \ln \left(1 + \frac{3}{t} \right) \right] = \lim_{t \rightarrow \infty} \frac{\ln(1 + 3/t)}{1/t}$$

2. **Apply L'Hôpital's Rule:** As $t \rightarrow \infty$, both the numerator and denominator approach 0. We differentiate both:

$$\lim_{t \rightarrow \infty} \frac{\frac{d}{dt}(\ln(1 + 3t^{-1}))}{\frac{d}{dt}(t^{-1})} = \lim_{t \rightarrow \infty} \frac{\frac{1}{1+3/t} \cdot (-3t^{-2})}{-1t^{-2}}$$

3. **Simplify the Expression:** The t^{-2} (or $1/t^2$) terms cancel out:

$$= \lim_{t \rightarrow \infty} \frac{\frac{1}{1+3/t} \cdot (-3)}{-1} = \lim_{t \rightarrow \infty} \frac{3}{1+3/t}$$

4. **Evaluate the Limit:** As $t \rightarrow \infty$, the term $3/t$ approaches 0.

$$\frac{3}{1+0} = 3$$

5. **Conclusion:** The limit is 3.

Solution to Example 21

The limit is evaluated at $x \rightarrow 0$. Because the definition of $|x|$ changes at $x = 0$, we must evaluate the left-hand limit ($x \rightarrow 0^-$) and the right-hand limit ($x \rightarrow 0^+$) separately.

1. **Analyze the Absolute Values near $x = 0$:**

- For $|3x + 4|$: As $x \rightarrow 0$, the expression $3x + 4$ approaches 4, which is positive. Therefore, for x near 0 (both from the left and right), $|3x + 4| = 3x + 4$.
- For $|x|$: The definition changes.
 - If $x \rightarrow 0^+$ (from the right), x is positive, so $|x| = x$.
 - If $x \rightarrow 0^-$ (from the left), x is negative, so $|x| = -x$.

2. **Calculate the Right-Hand Limit ($x \rightarrow 0^+$):** We use $|3x + 4| = 3x + 4$ and $|x| = x$.

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{(3x + 4) - (x) - 4}{x} &= \lim_{x \rightarrow 0^+} \frac{3x + 4 - x - 4}{x} \\ &= \lim_{x \rightarrow 0^+} \frac{2x}{x} = \lim_{x \rightarrow 0^+} 2 = 2 \end{aligned}$$

3. **Calculate the Left-Hand Limit ($x \rightarrow 0^-$):** We use $|3x + 4| = 3x + 4$ and $|x| = -x$.

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{(3x + 4) - (-x) - 4}{x} &= \lim_{x \rightarrow 0^-} \frac{3x + 4 + x - 4}{x} \\ &= \lim_{x \rightarrow 0^-} \frac{4x}{x} = \lim_{x \rightarrow 0^-} 4 = 4 \end{aligned}$$

4. **Conclusion:** The right-hand limit (2) is not equal to the left-hand limit (4). Since the one-sided limits do not match, the two-sided limit does not exist (DNE).